MIL-P-63142A (AR)
AMENDMENT 7
30 August 1985
SUPERSEDING
AMENDMENT 6
23 SEPTEMBER 1983

MILITARY SPECIFICATION

PROJECTILE, 8 INCH, HE, RA, M650
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-P-63142A (AR), dated 8 February 1981, and is approved for use by the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1, STANDARDS, MILITARY, add the following:

"MIL-STD-410 - Nondestructive Testing Personnel Qualification and Certification."

PAGE 2

* 2.1, Drawings, delete: "9288023 - Pellet Explosive"

3.2.2, line 1, delete: "grease or silicone" and substitute "lubricant".

Add the following new paragraph:

"3.2.4 Rotating band. The surface of the rotating band lands shall be examined for surface damage.

On the forward two lands, there shall be no evidence of raised copper due to nicks, digs or damage of any kind.

On the aft, three lands, any nicks or digs extending 1/2 the land width or less are acceptable, provided the maximum band diameter is not exceeded.

Surface scratches without evidence of raised copper are acceptable on all five lands.

FSC 1320

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PAGE 5

* 3.3.2: Delete "pellet" and substitute "booster".

3.3.5, line 4, delete: "4.4.2.9" and substitute "4.5.4."

Add paragraphs 3.3.6 and 3.3.7 as follows:

"3.3.6 Metal parts defects. There shall be no crack, pipe, or thin wall section in the aft warhead.

3.3.7 Foreign material. There shall be no foreign material in the explosive charge."

PAGE 6

Add the following new paragraph:

"3.4.7 Interior metal defects. The aft warhead cavity shall be free from embedded foreign matter, raised material or sharp edges."

PAGE 7

3.6.1.2.2, add the following: "unless the rocket motor fails to ignite."

Delete paragraph 3.6.2 in its entirety and substitute the following:

"3.6.2 Safety. There shall be no premature burst or indication of metal parts separation in the gun bore, or in flight, when fired at any of the conditions specified in 4.4.3.4. In addition, no projectile fired in the safety phase shall have a horizontal range less than 97.5 percent of the mean unless the rocket motor fails to ignite."

PAGE 8

3.10, line 2, delete: "in the exterior of the projectile" and substitute "in the nose threads or on the exterior of the aft warhead."

PAGE 9

* 4.3.1: Delete "Pellet, Explosive 9288023 10".

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PAGE 10

* Table II, delete: "Pellet, Explosive 4.4.2.7".

PAGE 11

* 4.4.1: Delete "h. Explosive.....manufacturer." and substitute:

"h. Booster assemblies PA-524 from one lot interfix from one manufacturer."

PAGE 12

4.4.2.1, change title to read: "Delay Assy and Motor Body Prior to Assembly."

Change Drawing Number block to read: "9287970/9288025".

Major 103, delete in its entirety and substitute the following:

Major category:

"Major 103: Rocket motor body protective cap missing or improper 0.40% 3.2 Visual."

Add the following Major defect:

Major category:

"Major 104: Nozzle loose or missing 100% 3.2 Visual/Manual".

PAGE 14

4.4.2.3, Major 103, delete: "Grease or silicone" and substitute "Lubricant".

Major 104, delete: "Silicone compound" and substitute "Lubricant"

PAGE 15

4.4.2.4, Major 101, delete: "Test" and substitute "Visual".

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4.4.2.5, delete "Critical 1: Nozzle insert missing, loose, inverted, or damaged" and substitute the following:

"1. Nozzle insert missing, or damaged, after assembly of delay".

PAGE 18

* Delete 4.4.2.7 in its entirety.

PAGE 19

4.4.2.8, Critical defect 2, delete Requirement Paragraph "3.4.6" and substitute "3.4.5".

Critical defect 3, delete Requirement Paragraph "3.4.8" and substitute "3.4.7".

Major 102 and 103: Delete in their entirety and substitute the following:

Major category:

"Major 102: Forward protective cap missing or improper 0.40% 3.2 Visual.

Major 103: Aft protective cap missing or improper 0.40% 3.2 Visual".

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* 4.4.2.9, Critical defect 3, delete Requirement Paragraph "3.4.7" and substitute "3.4.6".

Add the following critical defects:

"4. Presence of crack, pipe or thin wall section in the aft warhead 100% 3.3.6 4.4.3.1.2.

5. Foreign material in explosive charge 100% 3.3.7 4.4.3.1.3".

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Major category:

* Major 102, Inspection Method column: Delete "Gage" and substitute "Visual".

* Major 103: Delete "Explosive pellet" and substitute "Booster".

* Major 104: Delete in its entirety.

* Major 105: Delete in its entirety.

* Add the following Major defects:

Major category:

"Major 107: Booster cavity depth incorrect 0.40% 3.2 Gage.

Major 108: Booster cavity diameter incorrect 0.40% 3.2 Gage".

PAGE 22

* 4.4.2.10, delete: "Explosive pellet" and substitute "Booster".

PAGE 23

* 4.4.2.11, delete: "Disc" from title and substitute "Disk and Disc Assembly".

* Add Major defects 102 and 103 as follows:

Major category:

"Major 102: Adhesive missing or insufficient on disk (disc) prior to assembly of disc (disk) 0.40% 3.2 Visual

Major 103: Disc not assembled concentric to disk (allowing improper seating in aft warhead) 100% 3.2 Visual/Templet".

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* 4.4.2.12, Major 102, delete: "Disc" and substitute "Disk, disc".

PAGE 25

4.4.2.13, Major defect 105, delete: "15" in the No. of Sample Units block.

Major defect 106, delete "15" in the No. of Sample Units block.

Add "0.40%" to the AQL block.

PAGE 26

4.4.2.14, Critical defect 1: Delete "O-ring groove".

Major 101: Delete in its entirety.

Major defect 106: Delete in its entirety and substitute the following:

Major category:

"Major 106: Forward grain cushion improper, missing or improperly lubricated, prior to application of sealant 0.40% 3.2 Visual"

Major defect 108: Delete in its entirety and substitute the following:

Major category:

"Major 108: Aft protective cap missing or improper on aft warhead 0.40% 3.2 Visual."

PAGE 27

4.4.2.15, Major 103, delete: "Manual/Visual" in the Inspection Method column and substitute "Gage".

Major 105, delete "15" in the No. of Sample Units column.

Delete "4.3.3.2" in the Paragraph Reference/Inspection Method column and substitute "4.4.3.2".

Major 106, delete "3.2" in the Requirement paragraph and substitute "3.2.4".

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Major 107, delete in its entirety and substitute the following:

Major category:

"Major 107: Diameter of bourrelet, max. (applies if warhead of motor body sections have been retouched on OD) 100% 3.8 Gage."

PAGE 28

4.4.2.15, add the following two Minor defects:

Minor category:

"Minor 205: Lifting plug improperly lubricated 0.65% 3.2 Visual.

Minor 206: Warhead lot number not stamped on motor body rotating band 0.65% 3.2 Visual".

PAGE 29

4.4.2.16: Major defect 103, delete in its entirely and substitute the following:.

Major category:

"Major 103: Strapping or staples missing or loose 0.40% 3.2 Visual".

Minor Defect 203: Delete in its entirety and substitute the following:

Minor category:

"Minor 203: Strapping or staples improperly engaged 0.65% 3.2 Visual".

PAGE 31

Add 4.4.3.1.2 and 4.4.3.1.3 as follows:

"4.4.3.1.2 Metal parts defects. All radiographs of warhead section loading assemblies shall be examined for evidence of crack, pipe and thin wall sections in the aft warhead. The presence of any such indications shall cause that assembly to be classed defective and removed from the lot. (See Section 6.14).

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4.4.3.1.3 Foreign material. All radiographs of warhead section loading assemblies shall be examined for foreign material in the explosive charge. The presence of any foreign material shall cause that assembly to be classed defective and removed from the lot."

4.4.3.2, delete in its entirety and substitute the following:

"4.4.3.2 Motor body - warhead assembly torque (see 3.5 - Major defect. The torque applied to the motor body-warhead joint shall be witnessed for each projectile during assembly to assure compliance with the applicable requirement. Any projectile improperly assembled shall be classed defective and removed from the lot. In addition, a minimum of one sample projectile shall be randomly selected from each hours production for a check torque application. If any projectile fails to comply with the torque requirements, as evidenced by relative motion at the joint, when tested in accordance with 4.5.2 then all projectiles assembled since the last acceptable sample shall be rejected.

As an alternate to the once an hour torque check sample, a verification check may be made on the torque assembly machine. In this case, the torque readout from the assembly machine shall be verified for transducer accuracy at the beginning and end of the operating shift and after each four-hour production interval. If the torque assembly machine fails the verification test, all the projectiles assembled during that production interval shall be rejected. In the event of two consecutive gap rejects, the torque assembly machine display shall be checked during operation on a subsequent assembly. If the indicated torque is in or below the lower 20% of the acceptance range, a verification check shall be made. If the machine fails the verification, all projectiles assembled since the last acceptable verification shall be rejected."

4.4.3.3d, delete in its entirety and substitute the following:

"d. Voids or missing increments in the delay column."

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4.4.3.4, Critical defect 2, add the following: "unless the rocket motor fails to ignite."

Delete Critical defect 3 in its entirety.

Add Major defect 102 as follows:

Major category:

"Major 102: Rocket motor fails to ignite."

PAGE 33

Add paragraph 4.4.3.4.2.3 as follows:

"4.4.3.4.2.3 Failure of a rocket motor to ignite during +145°F safety tests shall be considered a Major defect, and the lot shall be rejected. If a motor ignition failure occurs during -50°F safety tests, the lot shall be referred to the procuring agency for disposition (see 6.14)."

4.4.3.5, Delete in its entirety and substitute the following:

"4.4.3.5 "Ring gage inspection, major defect. Any retouching on the aft warhead or motor body will require 100% ring gaging."

4.4.3.6: Delete in its entirety.

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4.4.3.7, delete in its entirety and substitute the following:

"4.4.3.7 Forward warhead - Aft warhead assembly torque (see 3.5. The torque applied to the forward warhead - aft warhead joint shall be witnessed for each projectile during assembly to assure compliance with the applicable requirement. Any projectile improperly assembled shall be classed defective and removed from the lot. In addition, a minimum of one sample projectile shall be randomly selected from each hour's production for a check torque application. If any projectile fails to comply with the torque requirements, as evidenced by relative motion at the joint, when tested in accordance with 4.5.2 then all projectiles assembled since the last acceptable sample shall be rejected.

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As an alternate to the once an hour torque check sample, a verification check may be made on the torque assembly machine. In this case, the torque readout from the assembly machine shall be verified for transducer accuracy at the beginning and end of the operating shift and after each four-hour production interval. If the torque assembly machine fails the verification test, all the projectiles assembled during that production interval shall be rejected. In the event of two consecutive gap rejects, the torque assembly machine display shall be checked during operation on a subsequent assembly. If the indicated torque is in or below the lower 20% of the acceptance range, a verification check shall be made. If the machine fails the verification, all projectiles assembled since the last acceptable verification shall be rejected."

4.4.3.8, delete the following sentence: "Fifteen (15) projectiles shall be randomly selected for this test."

4.4.3.9, line 1, delete: "Eight projectiles" and substitute "Six aft warheads".

Line 3, delete: "Any projectile" and substitute "Any aft warhead".

PAGE 35

4.5.1.1, lines 25 and 26, delete: "unless identified as extraneous artifacts or metal parts irregularities" and substitute "unless identified as film or processing related or as acceptable metal parts irregularities."

PAGE 36

4.5.1.2, delete the last sentence: "Beam alignment will be established to optimize detection of gap at aft end of propellant grain."

PAGE 38

4.5.1.6, first sentence, delete: "projectile" and substitute "aft warhead."

Last sentence, delete: "of each" which follows the words "In addition, one x-ray".

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Add the following new paragraph:

"4.5.1.7 Operator qualifications. For operation of radiographic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For setup and calibration of radiographic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II."

4.5.2, line 5, delete: "rocket-off cap" and substitute "rocket cap".

PAGE 42

4.5.3.2: Change "-65°F" to read "-50°F", (two places).

4.5.4, lines 5 and 6, delete: "projectile" and substitute "aft warhead".

PAGE 43

4.5.5., delete in its entirety and substitute the following:

"4.5.5, Ring gage. After touch up or joint painting, the projectile shall be inspected by passing through a ring gage manufacturered to the maximum allowable diameter. Ring gage to pass over bourrelet and aft warhead motor body joint."

4.5.6, delete in its entirety and substitute the following:

"4.5.6 The exterior of the aft warhead shall be visually inspected for exudation. If no exudate is found, remove the forward protective plug and inspect for exudation visually."

* 6.3, delete: "ARRADCOM, Attn: DRDAR-QAR-I" and substitute "ARDC, Attn: DRSMC-QAR-I (D)".

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* 6.8, delete "ARRADCOM, DRDAR-QAR-Q, and DRDAR-LCU-CP" and substitute "ARDC, DRSMC-QAR-Q (D), and DRSMC-LCU-SA".

* 6.9, delete: "ARRADCOM, DRDAR-QAR-Q and substitute "ARDC and DRSMC-QAR-Q (D)".

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* 6.11: Delete in its entirety.

* 6.12, delete: "ARRADCOM" at the end of paragraph and substitute "ARDC".

Add paragraph 6.14 as follows:

"6.14 Rocket motor ignition failure at -50°F. Degradation of rocket motor ignition reliability is expected at temperatures below -25°F. When an ignition failure occurs at -50°F during acceptance testing, the procuring agency must evaluate the following prior to determining lot disposition:

a. The delay assembly shall have functioned properly as evidenced by review of radar data.

b. The failure shall not be related to a change in components or processes.

c. The effect on compatibility with the M753 NUC round must be considered. This may require a retest at -30°F.

d. The rate of failure should not be significantly different from prior experience."

Add the following new paragraph:

"6.15 The procuring activity will be notified of any projectile rejected for metal parts defects found upon inspection of radiographs. The rejected projectile will be held pending disposition by the procuring activity."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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